

20. PLANET DATABASE

AIM

To write a Prolog program to create a Planet Database and retrieve planet information using Prolog queries.

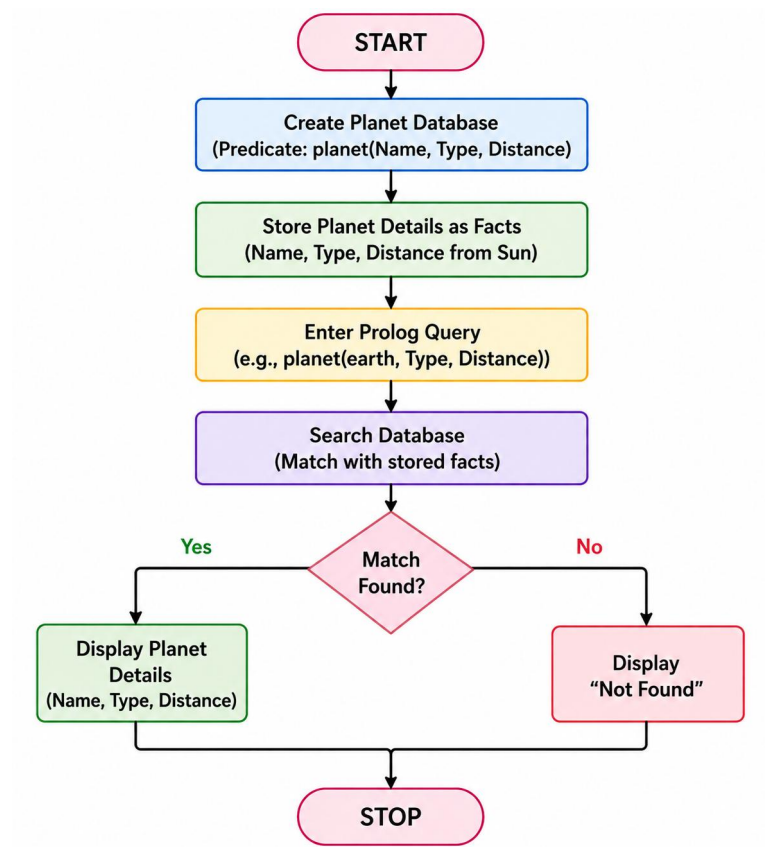
OBJECTIVE

- To create a database of planets using Prolog facts.
- To store the name, type, and distance of planets from the Sun.
- To retrieve planet details using queries.
- To understand database searching in Prolog.

ALGORITHM

1. Start.
2. Define the predicate planet(Name, Type, Distance).
3. Store the planet details as facts.
4. Enter a Prolog query.
5. Search and match the required information in the database.
6. Display the matching planet details.
7. Stop.

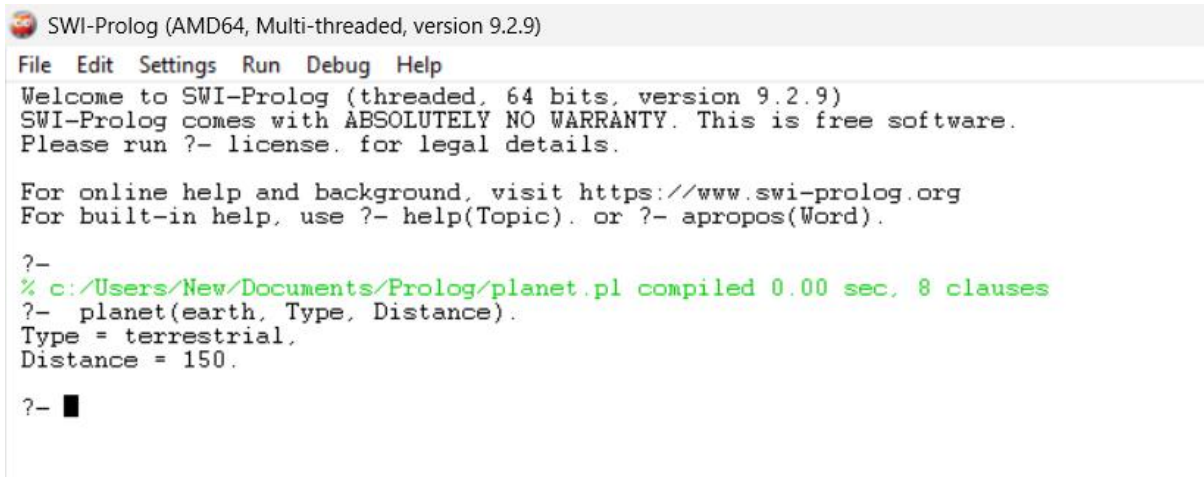
FLOWCHART:



CODE:

```
planet(mercury, terrestrial, 57).  
planet(venus, terrestrial, 108).  
planet(earth, terrestrial, 150).  
planet(mars, terrestrial, 228).  
planet(jupiter, gas_giant, 778).  
planet(saturn, gas_giant, 1430).  
planet(uranus, ice_giant, 2870).  
planet(neptune, ice_giant, 4495).
```

OUTPUT:

A screenshot of the SWI-Prolog (AMD64, Multi-threaded, version 9.2.9) window. The window has a menu bar with 'File', 'Edit', 'Settings', 'Run', 'Debug', and 'Help'. The main text area displays the following content: 'Welcome to SWI-Prolog (threaded, 64 bits, version 9.2.9)', 'SWI-Prolog comes with ABSOLUTELY NO WARRANTY. This is free software. Please run ?- license. for legal details.', 'For online help and background, visit https://www.swi-prolog.org', 'For built-in help, use ?- help(Topic). or ?- apropos(Word).', a prompt '?-', a green status line '% c:/Users/New/Documents/Prolog/planet.pl compiled 0.00 sec, 8 clauses', a query '?- planet(earth, Type, Distance).', the results 'Type = terrestrial,' and 'Distance = 150.', and a final prompt '?- ■'.

```
SWI-Prolog (AMD64, Multi-threaded, version 9.2.9)
File Edit Settings Run Debug Help
Welcome to SWI-Prolog (threaded, 64 bits, version 9.2.9)
SWI-Prolog comes with ABSOLUTELY NO WARRANTY. This is free software.
Please run ?- license. for legal details.

For online help and background, visit https://www.swi-prolog.org
For built-in help, use ?- help(Topic). or ?- apropos(Word).

?-
% c:/Users/New/Documents/Prolog/planet.pl compiled 0.00 sec, 8 clauses
?- planet(earth, Type, Distance).
Type = terrestrial,
Distance = 150.

?- ■
```

RESULT:

Thus, the Prolog Planet Database was successfully created, and the required planet information was retrieved using Prolog queries.